# PROOFLINE

Evidence-Backed Brand Governance — Final Product Submission

| Submission Item | Verified Detail / URL |
|---|---|
| Live Web Application | **https://proofline-4cz.pages.dev** (Cloudflare Pages) |
| Live Edge API Worker | **https://proofline-api.potlasrisharanchowdary.workers.dev/health** |
| Source Code Repository | **https://github.com/potlasrisharan/proofline** |
| Team Name & Lead | **Binary Brains** · Potla Sri Sharan (potlasrisharanchowdary@gmail.com) |
| Evaluation Window | Round 3: Oct 31, 11:59 PM – Nov 3, 11:59 PM (72 Hours) |
| Event Form | https://forms.gle/FxhT5Z1NupvGTGSw6 |

## 1. Executive Summary & Problem Formulation

Generative AI has increased marketing content production velocity by 10x, but brand safety and spec accuracy remain bottlenecks. Marketing teams frequently draft content with contradicted specs (e.g. 20h battery instead of approved 12h), expired discounts, unverified superlatives ('our most reliable laptop'), and missing legal qualifiers. Traditional brand guidelines sit in static PDFs that creators and AI tools never check.

**Proofline** is an evidence-first brand governance workspace operating as an automated pre-publication quality gate. Every claim is extracted, cross-checked against a versioned Brand Truth Registry, and linked to exact source evidence. Unlike generative tools that hallucinate confidence, Proofline operates on an **Evidence-or-Abstain** philosophy: if approved evidence does not exist, Proofline explicitly flags the claim as *Needs Human Review* rather than inventing facts.

## 2. The Three Improvements Delivered (Round 2 Commitments)

In Round 2, Team Binary Brains committed to delivering three exact improvements in Round 3. All three are fully built, tested, and demonstrated:

### ■ Improvement 1: Full Fact-Change Impact Radar (/impact)

• **Problem Solved:** When a product spec changes or a seasonal promotion ends, previously approved content drafts become outdated silently.
• **Delivered Solution:** An automated impact queue that maps claim-to-fact lineage across all registered content. When a spec changes, the Radar immediately identifies affected drafts, highlights old vs. new values (e.g. 10h → 12h battery life), flags records as re_review_required, and executes 1-click batch re-auditing with auditable task receipts.
• **Interactive Showcase:** The judge can simulate real-time spec updates directly on /impact and watch the queue dynamically revalidate.

## ■ Improvement 2: Evidence-or-Abstain Reliability Layer (/findings, API Verifier, Eval Suite)

• **Problem Solved:** AI verification tools lose enterprise trust when they hallucinate sources or claim an unverified statement is false.
• **Delivered Solution:** A strict tri-state verification engine distinguishing **Contradicted** (direct spec conflict), **Unsupported / Gap** (unsubstantiated claims where Proofline strictly abstains from guessing), and **Needs Human Review** (subjective tone). Backed by a high-severity deterministic verifier for exact units, currencies, and dates.
• **Benchmark Test Corpus:** 194-case VoltEdge evaluation dataset achieving **100% precision** on legal, offer, and promotional categories (tracked in docs/EVAL_REPORT.md).

## ■ Improvement 3: Governance Insights & Guideline-Gap Detector (/insights)

• **Problem Solved:** Repeated mistakes in marketing copy indicate ambiguous guidelines or training gaps, which corrections alone do not resolve.
• **Delivered Solution:** Telemetry dashboard tracking recurring mistake patterns by category, rule frequency, UTC weekly trends, and team/agency breakdowns (excluding toxic individual rankings).
• **Guideline-Gap Detector:** Analyzes frequent reviewer overrides to propose clearer guideline wording. Authorised brand owners can approve proposed wording updates with 1-click directly in the UI.

## 3. Technical Architecture & Implementation Summary

Proofline is built on modern, edge-native, decoupled infrastructure designed for high throughput, sub-50ms latency, and absolute data integrity:

| Layer | Technology Stack | Implementation Details & Scope |
|---|---|---|
| Frontend UI | Next.js 16 (App Router)<br>React 19 + TypeScript (Strict) | Responsive dashboard, side-by-side visual diffs, 1-click correction editor, theme contrast (WCAG AA), zero hydration errors. |
| Edge Deployment | Cloudflare Pages<br>(Global CDN) | Static HTML export deployed to 300+ edge cities. High availability, asset caching, sub-50ms TTFB worldwide. |
| Backend Edge API | Cloudflare Python Workers<br>(FastAPI ASGI Handler) | Deterministic verification engine, claim extractor, impact routes, and CORS middleware running edge compute. |
| Persistence & DB | Cloudflare D1 Database<br>(Remote SQLite) | Schema-versioned tables for Brand Truth Registry, claim links, content versions, and SHA-256 audit events. |
| Cross-Lingual AI | Sarvam AI Integration<br>(Telugu & Hindi) | Prompt-engineered Indic translation and terminology consistency auditor for Indian regional campaigns. |
| Gated Dispatch | Zernio Sandbox Mock<br>(Multi-Platform) | Simulated publishing to LinkedIn, Instagram, X, and Webflow unlocked only after human compliance approval. |
| Audit Integrity | SHA-256 Hash Chain | Cryptographic tamper-evident ledger validating sequential edits, suggestions, dismissals, and approvals. |

> ***Architectural Discipline:*** *Proofline strictly preserves system boundaries. The app provides a visible* ***Engine Switcher****: judges can execute against the live Cloudflare Python Worker & D1 database or test client-side via the Local Demo Engine with zero external network dependencies.*

## 4. Testing & Reliability Details ('Holds Up Without You')

Proofline has been stress-tested across edge cases so any judge can explore without encountering broken states, crashes, or hangs:

| Test Category | Count & Status | Verified Criteria & Failure-Mode Prevention |
|---|---|---|
| Playwright E2E Tests | **21 Tests Passed**<br>(1 skipped local-only) | Chromium/WebKit verification of judge 1-click flows, intake, diff rendering, Sarvam AI modal, and engine switching. |
| Backend Pytest Suite | **155 Tests Passed**<br>(0 failures) | Complete coverage of FastAPI routes, role RBAC, ingestion parsers (PDF, DOCX, images), and impact controllers. |
| VoltEdge Golden Dataset | **194 Cases Checked**<br>(100% precision) | 154/154 factual precision, unit conversions, currency parsing (USD/EUR/GBP/INR), prompt injection defence. |
| Type & Code Quality | **0 Errors / 0 Warnings** | TypeScript strict mode (tsc --noEmit) and ESLint 9 pass clean across all client and worker code. |

### Defensive UX & Fault Tolerance:

• **Empty Inputs:** Submitting blank or whitespace drafts produces clean inline alerts without 500 errors.
• **Direct Visit & Refresh:** Navigating directly to /findings with no active draft shows a friendly empty state with an instant **'■ Load Sample NovaBook S1 Review'** button.
• **Double-Submit Prevention:** All action buttons disable during processing (disabled={busy}) to eliminate race conditions.

## 5. Usability & Real-World Fit (Judge Walkthrough-Free Experience)

Proofline features an eye-catching **Judge Quick-Demo Header** right at the top of the workspace. Any judge can evaluate every major capability and all three Round 2 commitments in under 2 minutes:

| Preset Scenario | Target Route | What the Judge Experiences |
|---|---|---|
| ① **[IMP-1] Impact Radar** | /impact | Opens cascading impact queue, shows old vs new spec values, and runs 1-click batch re-audit. |
| ② **[IMP-2] Evidence-or-Abstain** | /?view=reviews | Highlights unverified claim ('our most reliable laptop') abstaining from guessing vs hard contradiction on battery. |
| ③ **[IMP-3] Guideline-Gap** | /insights | Displays recurring mistake telemetry and allows 1-click Brand Owner approval of clearer rule wording. |
| ④ **Claim Audit & Visual Diff** | /?view=reviews | Shows inline red strike-through flagged text and bold green suggested fix with 1-click apply. |
| ⑤ **Indic Verification (Sarvam)** | Modal | Cross-lingual Indian campaign brand checks across Telugu and Hindi powered by Sarvam AI. |
| ⑥ **1-Click Gated Dispatch** | Modal | Multi-platform publishing sandbox (LinkedIn, Instagram, X) unlocked only after review sign-off. |
| ⑦ **Cryptographic Audit Chain** | /?view=audit-log | Inspects SHA-256 tamper-evident hash ledger verifying sequential action integrity. |

## 6. Product Roadmap & Future Development Plan

Proofline has a concrete, phased path from this event MVP to an enterprise SaaS platform:

| Quarter | Focus Area | Key Deliverables & Milestones |
|---|---|---|
| **Q1 2027** | Ecosystem Plugins | • WordPress & Webflow native plugins for real-time claim validation in CMS editors.<br>• Google Docs Add-on & Figma plugin for pre-submission checks during drafting.<br>• Slack & MS Teams notification webhooks for high-risk approvals. |
| **Q2 2027** | Multi-Modal & Vision | • Multi-image brand palette distance, logo clearspace, and packaging compliance.<br>• Video caption/SRT auditing for YouTube, TikTok, and Instagram Reels.<br>• Cloudflare Workers AI OCR fallback for multi-language promotional posters. |
| **Q3 2027** | Regulatory Automation | • Automated ingestion of FTC, ASA, and ASCI advertising compliance rules.<br>• Multi-tenant workspace management for marketing agencies with 50+ clients.<br>• Automated periodic re-crawling of published web URLs against updated registries. |
| **Q4 2027** | Enterprise Scale | • SOC 2 Type II compliance & HIPAA-compliant redaction mode.<br>• Fine-tuned small language models deployed on edge workers for ultra-low latency.<br>• Enterprise SSO (Okta, Azure AD) with granular workspace role inheritance. |

***Team Binary Brains Verification:*** *This document, the live application, and source repository represent the complete, verified Round 3 submission. Built and delivered within the official event window.*
***Lead:*** *Potla Sri Sharan* · ***Live App:*** *https://proofline-4cz.pages.dev* · ***Repo:*** *https://github.com/potlasrisharan/proofline*